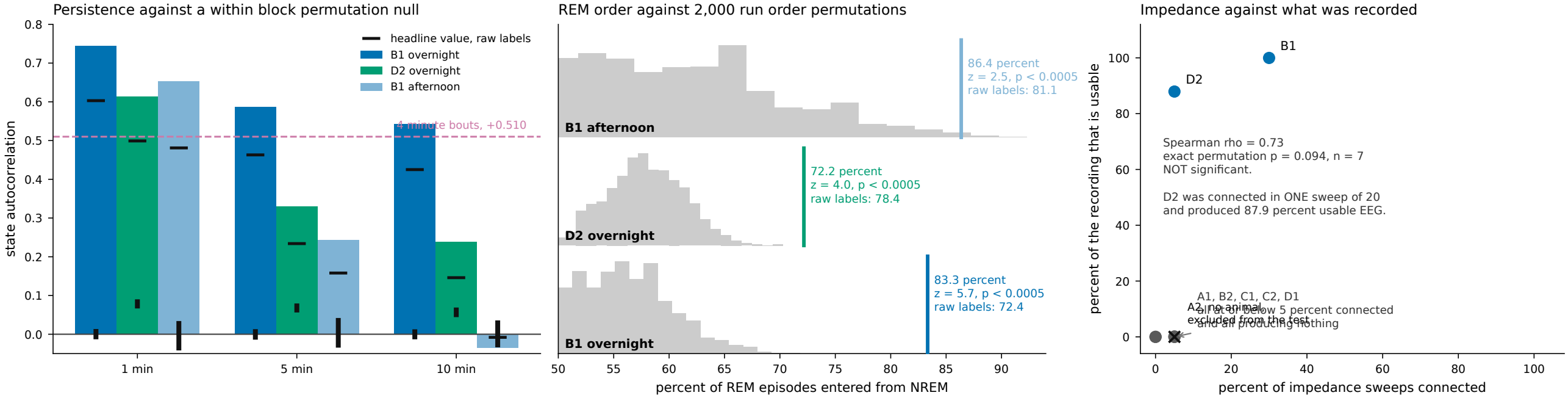
Figure 9. Statistics behind the sleep claims, and the limit of impedance as a predictor



LEFT. Autocorrelation of the asleep indicator, pooled across unbroken blocks. Bars come from the stored hypnogram, whose labels have been through the 12 second minimum bout rule; black dashes are the HEADLINE values, computed on RAW labels before that rule, and the gap between them is itself a finding, because smoothing inflates this statistic by construction. Thick black bars are the 95 percent range of 300 within block permutations. NO CONFIDENCE INTERVAL IS QUOTED ON THE MAGNITUDE, deliberately: the best recording is a single unbroken block, so there is no independent replicate to resample and a block bootstrap interval would largely report the chunk length chosen. A block bootstrap was built during this rebuild, produced intervals that did not bracket their own point estimate, and was discarded rather than tuned. MIDDLE. Run order permuted 2,000 times within blocks, preserving run lengths and composition. RIGHT. Rank correlation between impedance duty cycle and usable fraction across the seven positions that hold an animal, with an EXACT permutation p from 10,000 relabellings, because five of seven values are tied and the asymptotic p value is not trustworthy there. Seed 20260809 throughout.